Fine-Grained Species Recognition at Scale: Traditional Baselines, Transfer Learning, Robustness, and Grad-CAM

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Abstract

Fine-grained species recognition must separate visually similar taxa while remaining reliable across pose, scale, background, and image quality. We study this problem on a deterministic 500-species subset of iNaturalist 2021 Mini containing 20,000 training, 5,000 internal-validation, and 5,000 held-out test images. A split-controlled protocol compares handcrafted colour, texture, shape, and SIFT bag-of-visual-words pipelines with convolutional networks trained from scratch and with ImageNet transfer. The primary traditional model is a 3,704-D LinearSVC combining a 2,424-D handcrafted descriptor with 1,280-D spatial SIFT-BoVW, reaching 12.52% held-out top-1 accuracy. A pretrained EfficientNet-B0 reaches 72.74% top-1, 90.26% top-5, and 72.35% macro F1. The fixed 64-condition corruption study uses a distinct 2,424-D, 300-tree handcrafted Random Forest without BoVW as its traditional comparator and otherwise retains the frozen deep models. Confusion analysis and Grad-CAM expose fine-grained look-alikes as well as plausible colour and texture shortcuts. Together, the evidence establishes transfer learning as the strongest approach in this setting while identifying robustness, protocol-provenance, and explanation limitations that clean accuracy alone conceals.

1. Introduction

Species recognition is an unusually demanding image-classification problem. The label space is large, distinctions may depend on a small anatomical marking, and photographs are acquired opportunistically rather than under controlled laboratory conditions. In community-science collections, a target can occupy only a few pixels, be partially occluded, or resemble a closely related species while the background offers a tempting but unstable cue. The iNaturalist benchmarks capture these challenges at scale and have therefore become important tests of visual representation learning [29, 30].

This work asks how far progressively more expressive computer-vision methods can address a fixed 500-class recognition task. We begin with interpretable handcrafted representations—colour histograms, local binary patterns (LBP), histograms of oriented gradients (HOG), and SIFT bag-of-visual-words (BoVW)—paired with linear or tree-based classifiers. We then compare ResNet18 trained from random initialization against ImageNet-pretrained ResNet18 and EfficientNet-B0 models. Beyond clean-set accuracy, we test controlled image corruptions and inspect decisions with Grad-CAM. This broad comparison is useful because a single accuracy number cannot reveal whether a model is learning transferable morphology, memorising texture, or failing under a realistic loss of image quality.

Equally important is evaluation discipline. The official iNaturalist validation archive is used only as a held-out test source. Every fitted transform, visual vocabulary, classifier, and neural-network update sees the training partition alone. Hyperparameters, early stopping, and ablations use an internal validation partition. Stage 3 records its selected configurations before test access. For the rewritten Stage 2 track, the present selection manifest cryptographically freezes the complete selected state going forward, but the available history cannot independently establish that an equivalent lock existed before the already-recorded test evaluation. We therefore do not claim a retrospectively untouched Stage 2 test split. Figure 1 summarises the intended protocol.

Our contributions are fourfold. First, we construct a deterministic, audited 500-species protocol with 40 training, 10 validation, and 10 test images per species. Second, we provide complete traditional and deep baselines under common metrics, including training and inference timing. Third, we compare feature components, ImageNet pretraining, augmentation strength, and backbone freezing on validation data; the matched ResNet18 augmentation pair provides the strict one-factor ablation. Fourth, we complement clean evaluation with corruption robustness, confusion analysis, failure cases, and target-specific Grad-CAM visualisations. The result is not merely a leaderboard: it is evidence about which

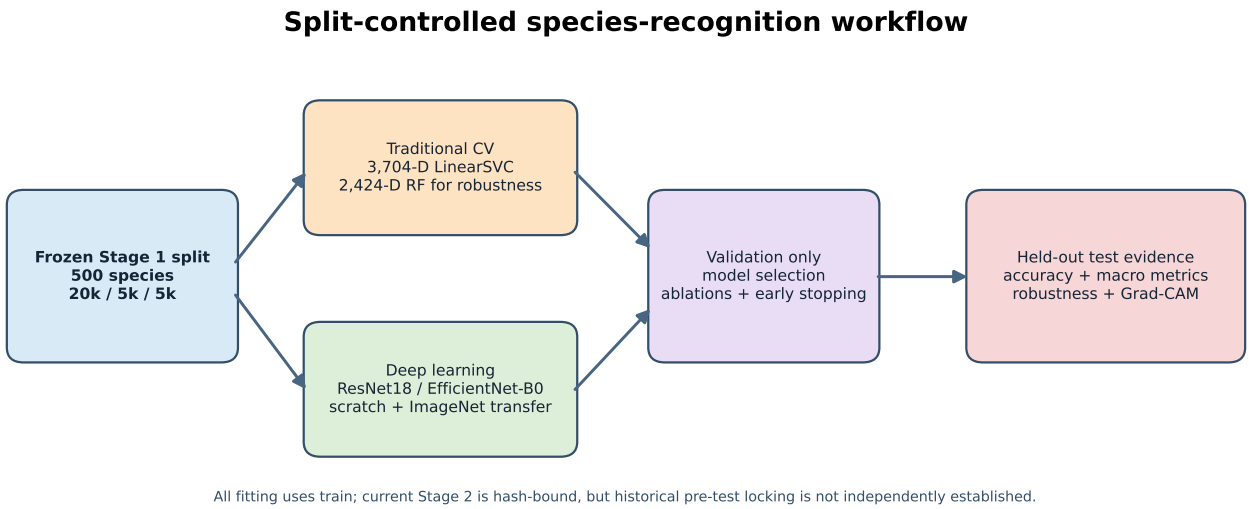


Figure 1. End-to-end project workflow. Traditional and deep models share the same frozen class mapping and partitions. Validation controls model/configuration selection and train-only pilots control feasibility; current artifacts are hash-bound before any future test access.

design choices matter, where performance breaks down, and which apparent explanations remain uncertain.

2. Literature Review

Fine-grained recognition and natural-world data. Fine-grained visual categorisation differs from generic object recognition because between-class variation can be smaller than within-class variation. Pose, life stage, sex, season, and viewpoint may transform an organism's appearance, while sibling species differ only in local structure. The iNaturalist 2018 dataset established a large, naturally imbalanced benchmark for species classification and detection [29]. The 2021 benchmark expanded this setting and highlighted the importance of representations that transfer across the long-tailed visual world [30]. Our balanced subset does not reproduce the full long tail, but retains the taxonomic diversity and uncontrolled image conditions that make the benchmark difficult.

Handcrafted representations. Before end-to-end representation learning, recognition pipelines separated feature design from classification. SIFT describes local gradient structure and is robust to scale and rotation changes [15]; BoVW quantises local descriptors into a fixed-length image histogram [6]. HOG aggregates oriented gradients over spatial cells and captures object shape [8], whereas uniform LBP encodes local texture transitions [18]. Colour histograms add information these grayscale descriptors omit. Linear support vector machines learn a maximum-margin decision boundary [5], and random forests combine decorrelated decision trees [3]. These methods offer low-cost, interpretable reference points, but their fixed statistics cannot adapt to the hierarchy of parts, textures, and context needed for hundreds of species.

Deep transfer learning. ImageNet enabled supervised pre-training at a scale sufficient to learn broadly useful visual primitives [9]. Residual connections make deeper networks easier to optimise [11], while EfficientNet uses compound depth/width/resolution scaling to improve parameter efficiency [25]. Features learned on a source task can transfer to a related target, although transferability generally decreases with task and layer distance [31]. Data augmentation reduces overfitting by encoding nuisance invariances; random cropping, colour perturbation, and erasing are common choices [23, 32]. We test both full fine-tuning and a frozen-backbone linear-head regime, because the latter reveals how much of the target task is already linearly separable in ImageNet features.

Robustness and explanation. Standard test sets can conceal brittleness to blur, noise, and compression. Image-corruption benchmarks formalise this concern by evaluating fixed models under parameterised degradations [12]. Our study adopts the same principle, though with a project-specific corruption suite and severity schedule applied to every held-out image. Grad-CAM instead probes spatial evidence: gradients of a target score weight convolutional feature maps to produce a coarse localisation heatmap [22]. Such maps are useful diagnostic evidence, but they do not prove causal feature use or explanation faithfulness. We therefore interpret them jointly with persisted predictions, true-target maps, and confusion statistics rather than as standalone explanations.

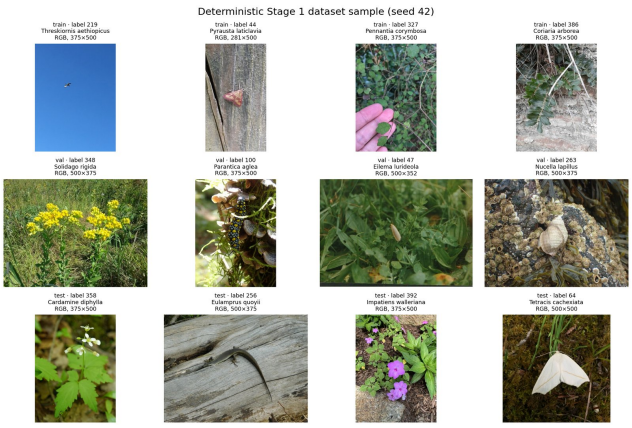


Figure 2. Examples from the frozen partitions. Images are EXIF-oriented for decoding; labels and assignments remain unchanged.

3. Dataset and Protocol

3.1. Frozen 500-species subset

We use iNaturalist 2021 Mini [30]. From the COCO-style annotations, a class is eligible only if it contains at least 50 images in `train_mini` and 10 images in the official validation archive. All 10,000 categories satisfy this threshold. We sort original category IDs, sample exactly 500 without replacement using NumPy PCG64 with seed 42, then sort the selected IDs and map them contiguously to labels 0–499. For each class, a class-specific deterministic permutation selects 50 `train_mini` images: 40 become training data and 10 become internal validation data. Ten images from the official validation archive form the held-out test partition.

The resulting manifests contain 20,000 training, 5,000 validation, and 5,000 test rows. Scientific/common names, original category IDs, image IDs, source splits, and portable local paths are retained. Automated validation checks exact per-class counts, contiguous labels, annotation provenance, safe paths, file existence, and duplicates by path, image ID, and UUID filename. No duplicate or cross-partition record was observed. Figure 2 shows deterministic samples and illustrates the substantial changes in object scale, viewpoint, background, and taxonomic kingdom.

3.2. Leakage controls and test isolation

All preprocessing that can learn from data is fitted on the training manifest only. This includes feature scaling, the SIFT vocabulary, linear and forest classifiers, and neural-network optimisation. Validation images determine early stopping, feature ablations, and the choice between pre-trained augmentation variants. Train-only pilots determine solver, memory, and runtime feasibility. The official test manifest is not read by training or validation commands. Stage 3 binds selected checkpoints and configurations to cryptographic hashes before test evaluation. The restored Stage 2 selection gate now enforces the same ordering for all future evaluations, although the repository cannot independently prove that this gate preceded the traditional test artifacts already present when the rewrite was integrated. Robustness and Grad-CAM operate only after this clean-test evaluation and never feed changes back into a model. This protocol prevents direct training-time image leakage and future iteration on held-out scores; the retrospective Stage 2 chronology limitation remains explicit.

Because the subset is class-balanced, overall accuracy and macro recall are numerically equal here, but we retain both concepts in the evaluation code. Balanced sampling also means every class contributes equally to macro metrics. It does not make the task easy: each species has only 40 learning examples, and the held-out per-class recall is quantised in increments of 0.1 because there are ten test examples per class.

4. Methods

4.1. Traditional computer vision

Block-structured handcrafted descriptor. After EXIF correction and RGB conversion, each image is resized bilinearly to 128×128. Three non-negative blocks are concatenated in the fixed order colour, texture, and HOG. Colour uses a joint hue–saturation histogram (12×8 bins) plus an eight-bin value histogram. Texture uses uniform LBP at $(P{=}8, R{=}1)$ and $(P{=}16, R{=}2)$. The colour and texture histograms are pooled over the whole image and a 2×2 grid, yielding 520 and 140 dimensions respectively. HOG uses nine orientations, 16×16-pixel cells, 2×2-cell blocks, square-root input, and L2-Hys normalisation, producing 1,764 values. The complete per-image handcrafted vector is therefore 2,424-D. Caches retain the raw blocks in manifest order; fitted normalisation remains inside each estimator pipeline.

SIFT–BoVW. We preserve aspect ratio while limiting the longest side to 256 pixels, convert to grayscale, and retain at most 300 SIFT keypoints per image [15]. A 256-word visual vocabulary is learned by mini-batch k-means from at most 100 descriptors from four deterministic *training* images per class. Neither validation nor test descriptors contribute to vocabulary fitting. Each local descriptor uses top-3 soft assignment to its three nearest words with Gaussian weights whose bandwidth is the median nearest-centre distance on sampled training descriptors, reducing visual-word boundary ambiguity [28]. Counts are pooled over the whole image and a 2×2 spatial grid [14], so the BoVW block has $256\times5 = 1{,}280$ dimensions. Images without keypoints receive an all-zero block.

Normalisation and classifier selection. Each selected block is independently ℓ_1 normalised, raised element-wise to

a power α, and ℓ_2 normalised. At α=0.5 this is the Hellinger map [1]; more generally, power normalisation suppresses dominating histogram bins [21]. Among 19 recorded validation configurations, the selected primary pipeline uses α=0.3, block-wise normalisation, and all four blocks. Its 3,704-D input is the 2,424-D handcrafted descriptor plus 1,280-D BoVW, followed by `LinearSVC` (liblinear, C=1). The separate BoVW-only comparator applies `LinearSVC` to 1,280 dimensions. The Random Forest uses 300 trees, square-root feature subsampling, minimum leaf size 2, and only the 2,424-D colour–texture–HOG descriptor.

The primary Stage 2 traditional comparison is the 3,704-D LinearSVC. Stage 4 instead uses the 2,424-D, 300-tree handcrafted Random Forest without BoVW as its canonical traditional robustness comparator: the corruption evaluator recomputes per-image handcrafted descriptors and applies the fitted pipeline, but it does not independently reconstruct the training-fitted SIFT vocabulary required for BoVW. The two traditional roles are therefore deliberately distinct.

4.2. Deep networks

We evaluate ResNet18 [11] and EfficientNet-B0 [25]. Each classifier head is replaced by a 500-output linear layer. The scratch ResNet18 starts from random weights. Transfer experiments initialise the backbone with the corresponding ImageNet weights and randomly initialise the new head. A frozen-head ablation holds the entire pretrained backbone fixed and keeps its BatchNorm layers in evaluation mode; the remaining experiments fine-tune every parameter.

At evaluation time, images are EXIF-oriented, converted to RGB, bicubically resized to 256 pixels, centre-cropped to 224×224, converted to a tensor, and normalised by ImageNet channel statistics. Basic training uses a random resized crop (scale 0.7–1.0) and horizontal flip. Strong training widens the crop scale to 0.5–1.0 and adds moderate colour jitter, a $\pm 15^\circ$ affine/rotation perturbation, and random erasing with probability 0.25. We intentionally omit vertical flips and extreme crops that could change biologically meaningful orientation or remove diagnostic parts.

For parameters θ and a labelled mini-batch, all networks minimise multiclass cross entropy,

$$\mathcal{L}(\theta) = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{\exp z_{i,y_i}}{\sum_{c=0}^{499} \exp z_{i,c}}, \tag{1}$$

using AdamW, cosine learning-rate annealing, weight decay 10^{-4}, and gradient-norm clipping at 1.0. Effective batch size is 32. Scratch training uses learning rate 10^{-3} for at most 25 epochs; fully fine-tuned pretrained models use 10^{-4} for at most 15 epochs; the frozen-head run uses 10^{-3} for at most 10 epochs. Validation macro F1 controls checkpointing, with top-1, top-5, loss, and earlier epoch as deterministic tie-breakers.

4.3. Robustness analysis

Let m_0 be a model's clean metric and $m_{k,s}$ its metric for corruption type k at severity s. We report absolute drop $\Delta_{k,s} = m_0 - m_{k,s}$, relative drop $\Delta_{k,s}/m_0$, mean corrupted performance across 16 conditions, and the worst observed condition. The fixed suite contains Gaussian blur, Gaussian noise, JPEG compression, and motion blur, each at four increasing severities. Corruptions are deterministic and applied to all 5,000 test images; model parameters and normalisation remain unchanged. This isolates sensitivity to acquisition quality from retraining or test-time adaptation.

4.4. Grad-CAM

For target class c, Grad-CAM weights convolutional channel k by the spatially averaged target-score gradient,

$$\alpha_k^c = \frac{1}{Z}\sum_{u,v} \frac{\partial z_c}{\partial A_{uv}^k}, \qquad L^c = \mathrm{ReLU}\left(\sum_k \alpha_k^c A^k\right). \tag{2}$$

We compute maps at EfficientNet-B0 layer `features.8`, upsample the activation map to the model crop, and align it back to the full image using the recorded centre-crop geometry. The analysis set is selected deterministically from persisted predictions: 12 high-confidence correct cases, 12 high-confidence errors, and nine representatives of top bidirectional confusion pairs (33 images, 54 target maps). Correct cases use the predicted target; errors include both predicted- and true-class targets. This target-specific pairing helps distinguish "where the predicted class scores" from "where the true class would score", but remains a diagnostic rather than a causal proof.

5. Experimental Setup

Implementation. Experiments run locally on an Apple M4 Pro. Deep models use PyTorch and torchvision [19] with MPS in FP32; traditional pipelines use scikit-learn [20] and OpenCV [2]. Data, image, configuration, progress, plotting, and serialisation utilities use NumPy, pandas, tqdm, scikit-image, Pillow, Matplotlib, joblib, torchvision, and PyYAML [4, 7, 10, 13, 16, 17, 24, 26, 27]. Random seeds for Python, NumPy, data samplers, workers, and model initialisation are fixed to 42. Data decoding applies EXIF orientation and handles grayscale or alpha-channel inputs by safe RGB conversion. Failed or unreadable images are fatal rather than silently skipped.

Scratch ResNet18 uses a 25-epoch cap and patience 6. The two pretrained ResNet18 augmentation variants share initial pretrained weights, head initialisation, sampler order, optimiser, scheduler, 15-epoch cap, and patience 4; only the training transform changes. EfficientNet-B0 uses the same strong-augmentation optimisation policy. A disposable preflight chooses a feasible micro-batch without contributing

weights or measurements to the canonical run. Before epoch one, the model is rebuilt and all random sources are reseeded.

Metrics. For each of 500 labels we compute precision, recall, and F1 with zero assigned when a denominator is empty, then report their unweighted macro averages. Top-k accuracy counts an example as correct when its true class is among the k largest scores. Score columns are aligned through the estimator's explicit class order, and stable descending ranking breaks exact ties in favour of the lower label. Numeric confusion matrices use rows for true labels and columns for predictions. We retain both overall/top-1 accuracy names for rubric compatibility, but they are the same statistic.

Traditional training time surrounds classifier `fit` only, and its test time surrounds scoring an already-loaded cached feature matrix. Deep training time includes loading, augmentation, transfer, forward/backward, and optimiser work in the training loop, excluding validation and serialisation. Deep test time is synchronised model-forward time; full evaluation wall time is also logged. Consequently, traditional and deep timing numbers describe different pipeline boundaries and should not be interpreted as a direct hardware-speed contest.

Selection and reproducibility. For Stage 3, validation metrics are independently recomputed from each best checkpoint before freeze. Test selection includes scratch ResNet18, the validation-best pretrained ResNet18 augmentation, and completed EfficientNet-B0; the basic-augmentation and frozen-head ResNet18 runs remain validation-only. For rewritten Stage 2, the selection manifest binds the 19 validation configurations, selected component order, normalisation, model and cache hashes, train-fitted BoVW vocabulary, and Stage 1 split hashes. Test evaluation rejects a missing or mismatched manifest, an unlisted model, and silent replacement of canonical metrics. This freezes the current state going forward. Because no independently verifiable pre-test manifest survives for the already-recorded traditional results, it does not establish historical chronology. Atomic caches, checkpoints, predictions, metric files, and configuration hashes nevertheless make the evidence currently reported traceable to its bound source. Figure 3 shows observed optimisation curves reconstructed from the persisted epoch logs, without rerunning inference.

6. Experimental Results

6.1. Clean validation and held-out test

Table 1 reports all completed baselines. The three traditional rows now represent distinct inputs. The primary 3,704-D colour–texture–HOG–BoVW LinearSVC reaches 12.52% test top-1, 26.78% top-5, and 11.16% macro F1. The 2,424-D, 300-tree handcrafted-only Random Forest without BoVW reaches 8.22%, 20.26%, and 6.54%, while the 1,280-D BoVW-only LinearSVC reaches 4.76%, 10.82%, and 4.14%. These values are smaller than the learned representations: scratch ResNet18 reaches 29.88% top-1, pretrained ResNet18 reaches 67.60%, and pretrained EfficientNet-B0 reaches 72.74% top-1, 90.26% top-5, and 72.35% macro F1. Validation/test rank agreement is descriptive only; it is not evidence that validation overfitting or prior test access did not occur.

The best network is also parameter-efficient: EfficientNet-B0 has 4.65M trainable parameters, versus 11.43M for ResNet18, yet improves test top-1 by 5.14 points over the selected pretrained ResNet18. Its canonical training loop takes 2,421.3 seconds, compared with 1,453.2 seconds for that ResNet18 and 2,772.5 seconds for scratch ResNet18. These times reflect different epoch counts and convergence behaviour, not architecture throughput in isolation. Figure 4 visualises the clean-score separation.

6.2. Validation-only ablations

Table 3 contains only validation results, so none of these comparisons consumes additional test information. The traditional rows show the selected full four-block validation reference beside colour-only, texture-only, and HOG-only controls. The complete 19-configuration grid also varied the power exponent, block-wise versus global normalisation, feature subsets, and the three canonical model roles. It selected block-wise α=0.3 with all four blocks; single-block scores must not be interpreted as a block's contribution in the combined model.

The matched ResNet18 augmentation ablation is the cleanest deep comparison: strong augmentation improves validation top-1 from 66.16% to 67.54% (+1.38 points) and macro F1 from 66.08% to 67.37% (+1.28 points). The scratch/basic run reaches 28.74% and the pretrained/basic run 66.16%; this large descriptive gap is consistent with the value of ImageNet transfer, but is not a one-factor causal estimate because the canonical learning rate, epoch cap, and stopping policy also differ. Similarly, the frozen-head configuration reaches 57.46% versus 66.16% for full fine-tuning, but uses a different optimisation schedule. It nevertheless shows that the pretrained representation is already useful before backbone adaptation.

6.3. Confusions and class-level failures

The normalised 500-by-500 confusion overview in Figure 5 is strongly diagonal, but local off-diagonal errors remain. The most frequent symmetric pair is Indian Pond Heron versus Squacco Heron, with six errors (two in one direction and four in the other). Several other leading pairs are taxonomically or visually close: Red-eyed Damselfly versus Common Bluetail, Yellow-faced versus Singing Honeyeater, and MacGillivray's versus Orange-crowned Warbler. Fig-

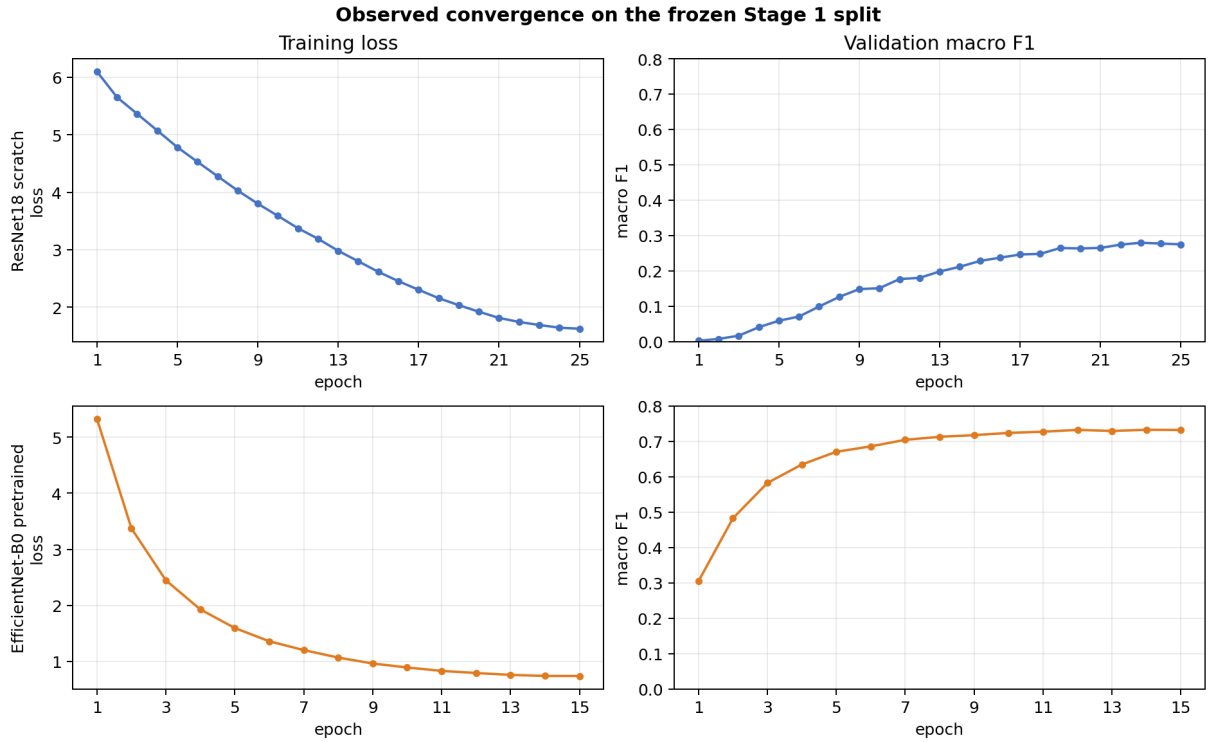


Figure 3. Training loss and validation macro F1 for scratch ResNet18 and the best pretrained EfficientNet-B0. Transfer learning converges faster and to a substantially stronger validation solution.

Table 1. Validation and held-out test results. All values are fractions and are derived from the current hash-bound evidence. The rewritten Stage 2 configuration is frozen going forward; its pre-test locking chronology cannot be independently established retrospectively.

Method	Pretr.	Val top-1	Val top-5	Val F1	Test top-1	Test top-5	Test F1
Handcrafted + BoVW (LinearSVC)	No	0.1294	0.2702	0.115401	0.1252	0.2678	0.111589
Random Forest	No	0.0860	0.2144	0.068372	0.0822	0.2026	0.065407
SIFT-BoVW (LinearSVC)	No	0.0472	0.1130	0.041271	0.0476	0.1082	0.041405
ResNet18 (scratch)	No	0.2874	0.5274	0.279552	0.2988	0.5364	0.292782
ResNet18 (ImageNet pretrained)	Yes	0.6754	0.8646	0.673652	0.6760	0.8624	0.672752
EfficientNet-B0 (ImageNet pretrained)	Yes	0.7368	0.9020	0.732628	0.7274	0.9026	0.723535

ure 6 preserves both error directions rather than collapsing them into a single uninterpretable count.

Class-level recall ranges from 0.1 for totara, Small-leaved Blinks, and New Zealand jasmine to 1.0 for multiple visually distinctive moth and beetle classes. These extremes should be interpreted cautiously because each recall uses only ten test images. High-confidence failures are more revealing. For example, a matrimony-vine image is predicted as Jerusalem cherry with 0.986 probability, and a Red Junglefowl is predicted as white pasqueflower with 0.932 probability. Such errors motivate the spatial analysis below.

6.4. Robustness to degradation

Table 4 and Figure 7 are generated only from a complete set of 16 corrupted conditions per model. EfficientNet-B0 averages 56.34% top-1 and 56.23% macro F1; its lowest recorded top-1 is 18.94%, and its lowest recorded macro F1 is 19.20%. These minima are computed independently and need not occur under the same corruption. The traditional curve is not the primary 3,704-D LinearSVC. It is the canonical 2,424-D, 300-tree handcrafted-only Random Forest comparator without BoVW, averaging 6.26% top-1 and 4.89% macro F1, with worst recorded top-1 and macro-F1 values of 2.36% and 1.45%, respectively; these minima are also computed independently. The build emits every value in this paragraph, the table, and the curves from the same

Table 2. Held-out macro metrics and measured fit/test timings. Overall accuracy equals top-1 for this single-label task; timing boundaries differ between cached-feature and end-to-end neural evaluation (Section 5).

Method	Macro precision	Macro recall	Macro F1	Train (s)	Test (s)
Handcrafted + BoVW (LinearSVC)	0.109007	0.125200	0.111589	1323.2	0.190
Random Forest	0.068343	0.082200	0.065407	870.5	1.370
SIFT-BoVW (LinearSVC)	0.038986	0.047600	0.041405	378.8	0.075
ResNet18 (scratch)	0.305342	0.298800	0.292782	2772.5	7.027
ResNet18 (ImageNet pretrained)	0.691092	0.676000	0.672752	1453.2	6.856
EfficientNet-B0 (ImageNet pretrained)	0.739866	0.727400	0.723535	2421.3	8.832

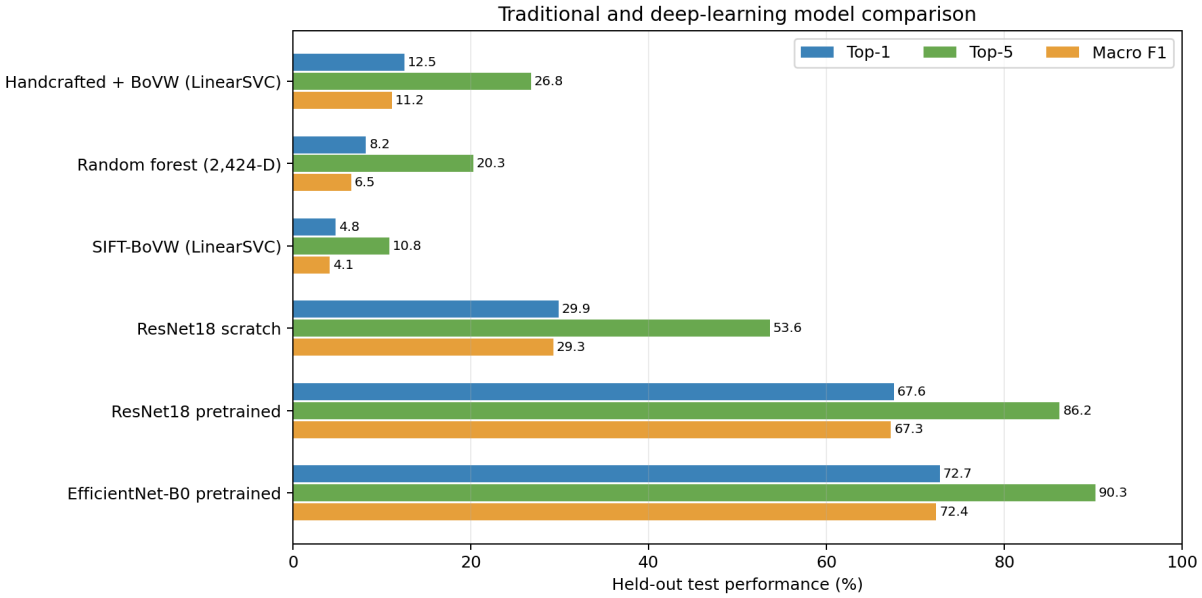


Figure 4. Held-out comparison of the six complete methods. The large gap between fixed descriptors and fine-tuned pretrained networks dominates the smaller within-family differences.

complete 64-cell evidence matrix after the Stage 2 model change.

The absolute results demonstrate that clean accuracy is not a robustness guarantee. The observed pretrained deep models retain higher clean and corrupted performance in the regenerated evidence, yet the gap between clean and worst-condition scores remains substantial. Robustness therefore remains a first-order failure mode rather than a minor correction to the clean ranking.

6.5. Grad-CAM and qualitative failure analysis

Figure 8 presents predicted-target overlays drawn only from the pre-registered analysis set. Correct predictions often concentrate on the organism: fruit clusters, flower heads, wing markings, or bird bodies. This localisation is encouraging but not sufficient evidence of the right fine-grained cue. The error cases are more diagnostic. Both matrimony vine and Jerusalem cherry maps emphasise red fruit, suggesting that shared colour and shape dominate subtler botanical distinctions. The Red Junglefowl map focuses on a bright textured region and supports a plausible texture shortcut for the flower prediction. For the shield-bug/tick error, the predicted map is tightly centred on the arthropod while the true-target map shifts, showing that correct localisation can coexist with incorrect classification.

Confused-pair maps likewise centre on the relevant animals, including the two herons and two damselflies. Their failure is therefore not simply background attention; it is consistent with insufficient resolution or feature specificity for closely related morphology. Conversely, some overlays span large rectangular crop regions, reflecting the coarse spatial resolution of the final convolutional layer. We consequently treat the maps as spatial associations rather than evidence of pixel-level causation.

Table 3. Validation-only traditional and deep-learning ablations.

Study membership	Experiment	Val top-1	Val top-5	Val macro-F1
Feature reference	All blocks + BoVW	0.1294	0.2702	0.115401
Feature ablation	Colour only	0.0608	0.1542	0.040555
Feature ablation	Texture (LBP) only	0.0356	0.1188	0.017688
Feature ablation	HOG only	0.0292	0.0852	0.019803
Pretraining	ResNet18 scratch / basic	0.2874	0.5274	0.279552
Pretrain / aug. / tuning ref.	ResNet18 pretrained / basic	0.6616	0.8538	0.660844
Augmentation	ResNet18 pretrained / strong	0.6754	0.8646	0.673652
Frozen head	ResNet18 frozen backbone	0.5746	0.7934	0.572458

Table 4. Fixed-protocol robustness summary over 16 corrupted test conditions per model. The traditional comparator is the 2,424-D handcrafted Random Forest, not the primary 3,704-D LinearSVC.

Model	Clean F1	Mean corrupt. F1	Worst F1	Mean Δ top-1	Max Δ top-1	Rank
EfficientNet-B0 pretrained	0.723535	0.562260	0.191962	0.164025	0.538000	1
ResNet18 pretrained	0.672752	0.478144	0.091338	0.198675	0.587800	2
ResNet18 scratch	0.292782	0.216528	0.077121	0.076512	0.211200	3
RF (2,424-D handcrafted)	0.065407	0.048932	0.014465	0.019550	0.058600	4

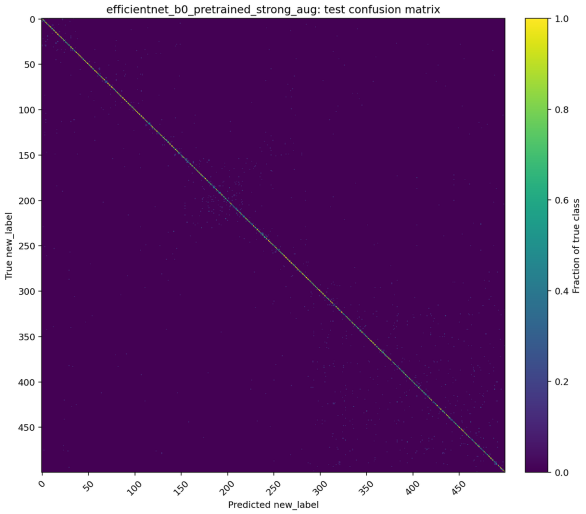


Figure 5. Row-normalised held-out confusion matrix for EfficientNet-B0 (rows true, columns predicted). The overview is complemented by the exact integer confused-pair counts in Figure 6.

7. Discussion

The clearest result is the importance of learned, transferred representations. Fixed colour/texture/gradient summaries provide a valid baseline but lose spatial and semantic structure; SIFT-BoVW adds local invariance without closing this gap. Scratch ResNet18 learns far stronger features from 40 examples per class, yet pretrained ResNet18 more than doubles its top-1 accuracy. EfficientNet-B0 then improves accuracy with fewer parameters than ResNet18, suggesting that its architecture and pretrained representation are well matched to the available compute and data. Because the scratch/pretrained schedules differ, this evidence is consistent with a pretraining benefit but does not isolate that benefit causally.

The traditional rewrite also sharpens what the classical comparison means. Joint colour statistics, multi-scale texture, HOG, and spatial soft-assignment BoVW are complementary enough that the 3,704-D LinearSVC is the primary Stage 2 traditional model. The Stage 4 Random Forest is a different, 2,424-D, 300-tree handcrafted-only comparator without BoVW. It remains useful for a fixed corruption study because that evaluator can deterministically recompute its per-image input and reuse the fitted normaliser, whereas it does not reconstruct the training-fitted visual vocabulary. Its robustness curve must therefore not be read as the robustness of the best-scoring traditional model.

The strict augmentation comparison is more modest but actionable. Strong augmentation improves the pretrained ResNet18 validation score without altering its initial weights, sampling order, optimiser, or stopping policy. The result is consistent with nuisance variability in iNaturalist images and supports the final use of colour, affine, crop, and erasing perturbations. However, the clean improvement does not make the model invariant to severe degradation. Motion blur removes high-frequency edges and local markings that may be diagnostic, plausibly contributing to its large top-1 fall. Training with blur or evaluating corruption-aware augmentation would be natural future work, but doing so after observing the held-out corruption scores would require a new validation protocol and test set.

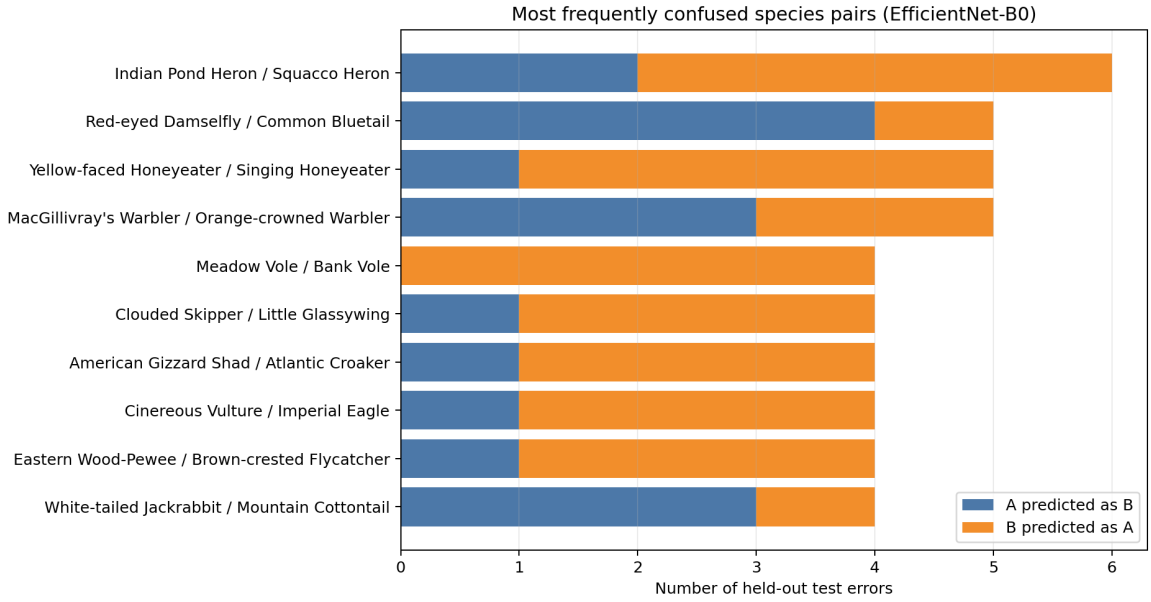


Figure 6. Ten most frequent bidirectional species confusions for EfficientNet-B0. Colour separates the two directions of each pair.

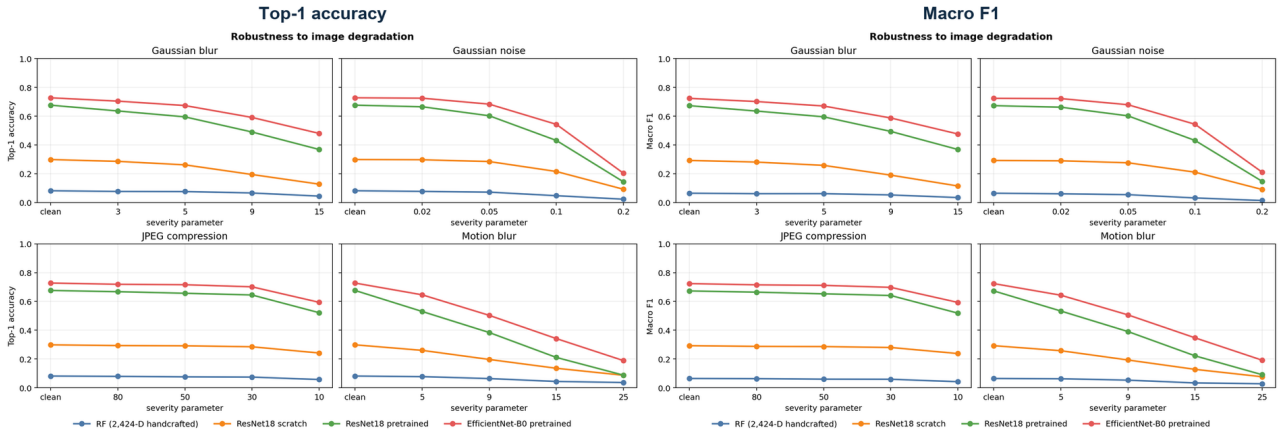


Figure 7. Top-1 and macro-F1 trajectories under four deterministic corruptions. Severity zero is the same frozen clean test evaluation for every curve. The traditional curve is the 2,424-D, 300-tree handcrafted Random Forest without BoVW, not the primary 3,704-D LinearSVC. No model is retrained or adapted.

Confusion and Grad-CAM analyses reveal two distinct error regimes. Some fine-grained pairs are correctly localised yet remain inseparable, indicating that global 224×224 classification may discard diagnostic detail. Other high-confidence failures align with conspicuous colour or texture, which may be predictive in training but not species-specific. A promising extension would combine higher-resolution or part-based attention with taxonomic hierarchy, while explicitly testing whether explanations remain stable under background and corruption interventions.

Several limitations bound our conclusions. The balanced 500-class subset is reproducible but does not measure full 10,000-class or long-tail performance. Forty training images per class constrain scratch learning, and ten test images per class make per-class metrics noisy. We evaluate one seed and a limited hyperparameter budget on one M4 Pro, so small model differences may not be statistically stable. Traditional cached-feature timing and deep forward timing have different scopes. The corruption suite covers four synthetic degradations, not distribution shifts in geography, season, camera, or observer behaviour. Finally, Grad-CAM is coarse and gradient-based; its maps are useful for forming hypotheses, not certifying faithful reasoning. A further provenance limitation is retrospective: the integrated repository contains

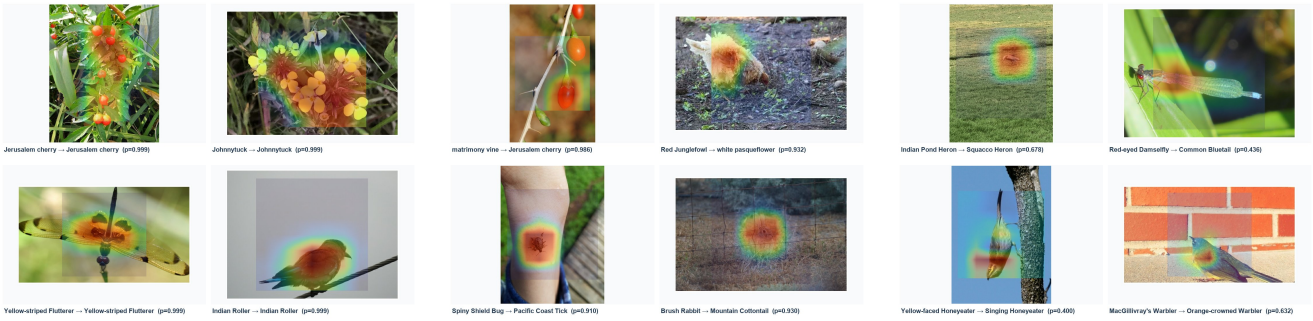


Figure 8. Persisted EfficientNet-B0 Grad-CAM overlays. Left: four high-confidence correct predictions. Centre: four high-confidence errors. Right: representatives of frequent confused pairs. Captions show true → predicted common names and softmax confidence. These are qualitative diagnostics, not causal explanations.

validation and test evidence for rewritten Stage 2, but no surviving artifact independently proves that the selected configuration was immutably locked before the first test access. The restored selection manifest freezes the current hashes, allowed models, and validation grid going forward; it does not manufacture a historical chronology.

Despite these limitations, the project has two practical strengths. First, the current split, model, prediction, metric, and selection artifacts are cross-hashed, so subsequent drift or unauthorised test access is detectable. Second, the combination of clean metrics, macro scores, classwise errors, corruptions, and explanations gives a more complete account than accuracy alone. The best model is genuinely useful on clean images, but the robustness and failure analyses show exactly why it should not yet be treated as a dependable field identifier without uncertainty handling and expert review.

8. Conclusion

We developed and evaluated a complete 500-species recognition pipeline on a deterministic iNaturalist subset. The rewritten primary traditional model combines colour, texture, HOG, and spatial SIFT-BoVW in a 3,704-D LinearSVC and reaches 12.52% held-out top-1. ResNet18 trained from scratch reaches 29.88%, while ImageNet transfer raises the best ResNet18 result to 67.60%. Pretrained EfficientNet-B0 is the strongest model at 72.74% top-1, 90.26% top-5, and 72.35% macro F1. A matched ablation finds a 1.38-point validation gain from stronger augmentation.

The advanced studies qualify that success. The corruption analysis uses a distinct 2,424-D, 300-tree handcrafted Random Forest without BoVW as the traditional comparator and derives every aggregate only after all 64 cells are regenerated. Confusion pairs concentrate among visually similar species, and Grad-CAM examples suggest both meaningful organism localisation and possible colour/texture shortcuts. The main lesson is therefore two-sided: transfer learning and efficient convolutional architecture provide a large practical advance, but reliability requires evaluation beyond the clean test set. The current selection manifest, persisted predictions, and reproducible analyses provide a sound basis for future higher-resolution, corruption-aware, and taxonomy-informed methods, while the retrospective Stage 2 locking limitation remains part of the evidence record.

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